01. Analyze the Complete Project First

Beginner-friendly requirements page

In simple words

This is the evidence-gathering stage. Read the real frontend, backend, Keycloak export, Docker files, environment examples, tests, and deployment files before making claims.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 01-analyze-the-complete-project-first.md

02. Project Overview

Beginner-friendly requirements page

In simple words

Describe the application in plain language and show how the user, application, Keycloak, tokens, and protected API fit together.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 02-project-overview.md

03. What Is Keycloak?

Beginner-friendly requirements page

In simple words

Explain Keycloak as an identity and access-management server that authenticates users, issues tokens, and centralizes access rules.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 03-what-is-keycloak.md

04. Why Use Keycloak?

Beginner-friendly requirements page

In simple words

Compare application-owned login with centralized identity management, including SSO, standard protocols, and reduced authentication code.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 04-why-use-keycloak.md

05. Complete Keycloak Architecture

Beginner-friendly requirements page

In simple words

Map browser, frontend, backend, Keycloak realm, client, database, and any application database to the actual deployment.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 05-complete-keycloak-architecture.md

06. Technology Stack

Beginner-friendly requirements page

In simple words

Inventory only technologies found in the repository and explain what each one does, where it runs, and how it communicates.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 06-technology-stack.md

07. Keycloak Components

Beginner-friendly requirements page

In simple words

Introduce realm, client, user, role, scope, mapper, identity provider, and authentication flow, giving detail only where the project uses them.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 07-keycloak-components.md

08. Realm

Beginner-friendly requirements page

In simple words

Explain a realm as an isolated identity space and document its real name, settings, users, clients, and roles without exposing secrets.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 08-realm.md

09. Client

Beginner-friendly requirements page

In simple words

Explain the Keycloak client as the registered application identity and document its ID, flow settings, redirect URIs, origins, and authentication mode.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 09-client.md

10. Client Types

Beginner-friendly requirements page

In simple words

Explain public, confidential, and bearer-only clients, then identify the project's type and the security reason for that choice.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 10-client-types.md

11. User

Beginner-friendly requirements page

In simple words

Explain how a Keycloak user is stored and authenticated, covering profile fields, enabled state, credentials, roles, and groups without passwords.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 11-user.md

12. Complete Login Flow

Beginner-friendly requirements page

In simple words

Trace the real click-to-login sequence from the frontend to the authorization endpoint, login page, callback, token endpoint, and authenticated state.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 12-complete-login-flow.md

13. OAuth 2.0 Explanation

Beginner-friendly requirements page

In simple words

Map resource owner, client, authorization server, resource server, authorization code, access token, and scope to this project.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 13-oauth-2-0-explanation.md

14. OpenID Connect

Beginner-friendly requirements page

In simple words

Explain OIDC as the identity layer on OAuth 2.0, especially the ID token, user identity, and UserInfo endpoint.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 14-openid-connect.md

15. Authorization Code Flow

Beginner-friendly requirements page

In simple words

Explain why a temporary authorization code is returned first and then exchanged for tokens by the correct client.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 15-authorization-code-flow.md

16. PKCE

Beginner-friendly requirements page

In simple words

Check whether PKCE is implemented. Explain verifier, challenge, and interception protection, and explicitly mark it as absent when not found.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 16-pkce.md

17. Token Types

Beginner-friendly requirements page

In simple words

Compare access, ID, and refresh tokens by purpose, recipient, lifetime, and safe handling.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 17-token-types.md

18. JWT Structure

Beginner-friendly requirements page

In simple words

Explain header, payload, and signature using a fake token and distinguish claims actually consumed by code from common Keycloak claims.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 18-jwt-structure.md

19. Complete Token Flow

Beginner-friendly requirements page

In simple words

Connect login, code exchange, token issuance, token storage, API authorization header, validation, expiry, and refresh.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 19-complete-token-flow.md

20. Protected API Flow

Beginner-friendly requirements page

In simple words

Trace a protected request from frontend call through bearer-token extraction, validation, authorization, business logic, and response.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 20-protected-api-flow.md

21. API Documentation

Beginner-friendly requirements page

In simple words

Document actual endpoints, methods, inputs, outputs, authentication requirements, roles, success responses, and failure responses.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 21-api-documentation.md

22. Authentication Middleware

Beginner-friendly requirements page

In simple words

Identify middleware that proves token authenticity and explain issuer, signature, expiration, audience, and claim checks.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 22-authentication-middleware.md

23. Authorization

Beginner-friendly requirements page

In simple words

Separate authentication from authorization and show how the project decides whether an authenticated user may perform an action.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 23-authorization.md

24. Realm Roles

Beginner-friendly requirements page

In simple words

Explain realm-wide roles and document only roles present in the realm export or configuration.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 24-realm-roles.md

25. Client Roles

Beginner-friendly requirements page

In simple words

Explain application-specific roles and show where their names are emitted, mapped, and checked.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 25-client-roles.md

26. Role Mapping

Beginner-friendly requirements page

In simple words

Trace how a user, group, realm role, or client role becomes an effective authorization decision.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 26-role-mapping.md

27. Groups

Beginner-friendly requirements page

In simple words

Check whether groups exist and explain inheritance and membership only when demonstrated by project configuration.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 27-groups.md

28. Client Scopes

Beginner-friendly requirements page

In simple words

Explain scopes as sets of requested permissions and mapped claims, distinguishing default and optional scopes.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 28-client-scopes.md

29. Protocol Mappers

Beginner-friendly requirements page

In simple words

Explain how mappers place identity or authorization data into tokens and identify every mapper relied on by the app.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 29-protocol-mappers.md

30. UserInfo

Beginner-friendly requirements page

In simple words

Document whether the application calls UserInfo and why it would use that endpoint instead of relying only on token claims.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 30-userinfo.md

31. Logout Flow

Beginner-friendly requirements page

In simple words

Trace browser logout, Keycloak session termination, redirect behavior, local token cleanup, and failure handling.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 31-logout-flow.md

32. Refresh Token Flow

Beginner-friendly requirements page

In simple words

Explain refresh-token exchange, rotation or reuse settings, expiry, revocation, and secure storage expectations.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 32-refresh-token-flow.md

33. Keycloak Admin Console

Beginner-friendly requirements page

In simple words

Explain the console areas needed to inspect realms, clients, users, roles, scopes, sessions, and events.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 33-keycloak-admin-console.md

34. Docker / Keycloak Startup

Beginner-friendly requirements page

In simple words

Trace container startup, environment loading, realm import, port exposure, health readiness, and dependency order.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 34-docker-keycloak-startup.md

35. Keycloak Database

Beginner-friendly requirements page

In simple words

Identify the configured database and explain which identity data belongs to Keycloak versus application business data.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 35-keycloak-database.md

36. Complete Database Flow

Beginner-friendly requirements page

In simple words

Show how configuration and identity records are read or written during startup, login, token issuance, and administration.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 36-complete-database-flow.md

37. Error Handling

Beginner-friendly requirements page

In simple words

For each failure, record cause, layer, status, response, user experience, and debugging evidence.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 37-error-handling.md

38. Common Keycloak Errors

Beginner-friendly requirements page

In simple words

Create Cause -> Check -> Fix guides for 401, 403, token, issuer, audience, redirect, client, realm, CORS, and connection errors.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 38-common-keycloak-errors.md

39. Security Analysis

Beginner-friendly requirements page

In simple words

Separate implemented controls from production recommendations for HTTPS, secrets, PKCE, CORS, CSRF, XSS, MFA, brute force, and rate limits.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 39-security-analysis.md

40. Environment Variables

Beginner-friendly requirements page

In simple words

List variable names, purpose, consumers, and safe example formats while redacting real secret values.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 40-environment-variables.md

41. Frontend Flow

Beginner-friendly requirements page

In simple words

Document login initiation, callback, token handling, auth state, protected routes, API calls, and logout using actual files and functions.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 41-frontend-flow.md

42. Backend Flow

Beginner-friendly requirements page

In simple words

Document configuration loading, route registration, middleware order, token validation, authorization, and business logic.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 42-backend-flow.md

43. Complete End-to-End Example

Beginner-friendly requirements page

In simple words

Walk through one login-to-protected-API scenario step by step, naming the request, response, file, function, token, and failure point.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 43-complete-end-to-end-example.md

44. Complete Token Example

Beginner-friendly requirements page

In simple words

Use only fake, sanitized JWT payloads to illustrate issuer, subject, audience, username, and expiry.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 44-complete-token-example.md

45. Keycloak vs Custom JWT

Beginner-friendly requirements page

In simple words

Compare who handles users, login, token issuance, roles, revocation, and operational burden.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 45-keycloak-vs-custom-jwt.md

46. Keycloak vs Other Identity Providers

Beginner-friendly requirements page

In simple words

Give a focused comparison with Auth0, Okta, Cognito, and Entra ID, centered on why this project may choose Keycloak.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 46-keycloak-vs-other-identity-providers.md

47. Scalability

Beginner-friendly requirements page

In simple words

Explain how the design changes from a local setup to thousands, hundreds of thousands, and millions of users.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 47-scalability.md

48. Production Keycloak Architecture

Beginner-friendly requirements page

In simple words

Describe load balancing, multiple Keycloak instances, a durable database, applications, and resource APIs.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 48-production-keycloak-architecture.md

49. High Availability

Beginner-friendly requirements page

In simple words

Explain redundancy, health checks, failover, backups, disaster recovery, and database availability.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 49-high-availability.md

50. Monitoring

Beginner-friendly requirements page

In simple words

Define useful production signals: login failures, token failures, health, latency, error rate, CPU, memory, and sessions.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 50-monitoring.md

51. Testing

Beginner-friendly requirements page

In simple words

Inventory existing tests and recommend authentication, authorization, token, API, logout, and error cases when missing.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 51-testing.md

52. Debugging Workflow

Beginner-friendly requirements page

In simple words

Give a repeatable browser -> redirect -> realm/client -> user -> backend -> token investigation path for login and API failures.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 52-debugging-workflow.md

53. Keycloak CLI / Admin Commands

Beginner-friendly requirements page

In simple words

Include only commands compatible with the detected Keycloak version and startup method.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 53-keycloak-cli-admin-commands.md

54. Folder Structure

Beginner-friendly requirements page

In simple words

Explain the real directories and identify where frontend, backend, Keycloak, Docker, tests, and documentation live.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 54-folder-structure.md

55. File-by-File Explanation

Beginner-friendly requirements page

In simple words

For every important file, capture purpose, imports, functions, callers, calls, auth responsibility, inputs, and outputs.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 55-file-by-file-explanation.md

56. Mermaid Diagrams

Beginner-friendly requirements page

In simple words

Create diagrams for architecture, realm, client, login, code flow, token validation, protected API, logout, roles, database, and production.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 56-mermaid-diagrams.md

57. Beginner Glossary

Beginner-friendly requirements page

In simple words

Define identity terms in one or two plain-language sentences, including realm, client, JWT, issuer, audience, SSO, and MFA.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 57-beginner-glossary.md

58. Common Beginner Confusions

Beginner-friendly requirements page

In simple words

Answer the common distinctions: Keycloak versus database, backend, API, JWT, OAuth, OIDC, access token, ID token, and frontend trust.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 58-common-beginner-confusions.md

59. Interview Questions

Beginner-friendly requirements page

In simple words

Create project-specific beginner, intermediate, and advanced questions with technically correct, easy-to-say answers.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 59-interview-questions.md

60. 2-Minute Project Explanation

Beginner-friendly requirements page

In simple words

Provide a short interview answer describing the actual application, Keycloak responsibilities, tokens, backend validation, and role checks.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 60-2-minute-project-explanation.md

61. 5-Minute Technical Explanation

Beginner-friendly requirements page

In simple words

Provide a deeper narrative from user and realm through client, authentication, code, tokens, API validation, roles, and response.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 61-5-minute-technical-explanation.md

62. Learning Path

Beginner-friendly requirements page

In simple words

Order learning from HTTP to authentication, JWT, OAuth, OIDC, Keycloak, RBAC, SSO, MFA, and production IAM.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 62-learning-path.md

63. Documentation Files

Beginner-friendly requirements page

In simple words

Define a maintainable docs folder, stable filenames, a master index, and links between topic pages.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 63-documentation-files.md

64. Master Markdown File

Beginner-friendly requirements page

In simple words

Assemble the complete learning path into one navigable Markdown document with implementation notes and production recommendations.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 64-master-markdown-file.md

65. Complete PDF

Beginner-friendly requirements page

In simple words

Create a polished PDF containing the complete documentation, diagrams, examples, tables, page numbers, and redacted examples.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 65-complete-pdf.md

66. PDF Quality Check

Beginner-friendly requirements page

In simple words

Verify clipping, diagram overlap, readable code, table width, page numbers, contents, JSON formatting, and secret redaction.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 66-pdf-quality-check.md

67. Final Master Flow

Beginner-friendly requirements page

In simple words

End with one compact end-to-end flow from user to application, realm, client, login, tokens, API, validation, authorization, and response.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 67-final-master-flow.md

68. Most Important Question

Beginner-friendly requirements page

In simple words

Answer exactly what happens after Login is clicked, including where, why, request, response, token, failure, and production equivalent.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 68-most-important-question.md

69. Accuracy Rules

Beginner-friendly requirements page

In simple words

Treat repository evidence as authoritative, never invent configuration, never expose secrets, and clearly separate implemented behavior from recommendations.

Required workflow

Inspect repository -> identify evidence -> explain simply -> trace request/response -> verify -> label missing features.

Evidence checklist

Capture	What the final documentation must contain
Implementation	Exact files, functions, configuration, and observed behavior
Workflow	Actors, request, response, status, token, and failure path
Safety	Fake examples only; redact secrets and real tokens
Clarity	Implemented behavior separated from production recommendations

Expected result

A beginner can explain the concept, follow the workflow, locate the implementation, and distinguish observed project behavior from recommended improvements.

Markdown page: 69-accuracy-rules.md